

Kaushlendra Kumar

Curriculum Vitae

School of Mathematical Sciences
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Profile

Research My research uses exact algebraic and geometric structures to extract concrete physics across mathematical physics. I work on exact Yang–Mills theory and geometric transport, Quantum Riemannian Geometry and finite spectral triples in noncommutative geometry, electromagnetic knots, matrix/emergent geometry, and quantum information geometry. The common aim is to compute geometry explicitly—connections, curvature, spectral distances, phases, charges, or channel metrics—and identify which physical structures are forced, constrained, or ruled out.

Positions

- 01 Mar, 2024 – 14 Nov, 2027 **DFG Walter-Benjamin Research Fellow**, *School of Mathematical Sciences*, Queen Mary University of London, Project with Shahn Majid on particle physics from Quantum Riemannian Geometry.
- 01 Jul, 2023 – 29 Feb, 2024 **Postdoc**, *Institute of Physics*, Humboldt University of Berlin, Germany. Funded by the Volkswagen Stiftung grant of the host, Nichol Furey.
- 2023 – 2024 (4 months) **Junior Research Fellow**, *Erwin Schrödinger Institute for Mathematics and Physics*, University of Vienna.
- 05 May, 2022 – 30 Jun, 2023 **Postdoc**, *Institute for Theoretical Physics*, Leibniz University Hannover (ITP-LUH), Germany. Funded by DAAD and ITP-LUH.

Education

- 01 Oct, 2018– 04 May, 2022 **PhD Theoretical Physics**, *Institute for Theoretical Physics*, Leibniz University Hannover, Germany.
Thesis title Solutions of Yang–Mills theory in four-dimensional de Sitter space
Grade Magna cum Laude
Defense date 4 May, 2022
- 01 Jun, 2018– 30 Sep, 2018 **German language**, *Carl Duisberg Centrum Berlin*, mandatory pre-PhD language course funded by DAAD, Level reached: B1 (Intermediate).
- 23 Jul, 2015– 31 May, 2018 **MSc Physics**, *Department of Physical Sciences*, Indian Institute of Science Education & Research, Kolkata (IISER-K), India.
Grade 9.25/10
Defense date 20 Jul, 2017
- 23 Jul, 2010– 08 May, 2015 **BTech & MTech Biochemical Engineering**, *School of Biochemical Engineering*, Indian Institute of Technology, Banaras Hindu University (IIT-BHU), Varanasi, India.
Grade 7.88/10
Defense date 27 Apr, 2015

List of publications

(a) Regular articles

- 2024 **Kaushlendra Kumar** and Harold C. Steinacker. Modified Einstein equations from the 1-loop effective action of the IKKT model. *Classical and Quantum Gravity*, volume 41, page 185007, 2024. DOI:10.1088/1361-6382/ad6e4b.
- 2024 Savan Hirpara, **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Exact gauge fields from anti-de Sitter space. *Journal of Mathematical Physics*, volume 65, page 072903, 2024. DOI:10.1063/5.0150027.
- 2022 **Kaushlendra Kumar**, Olaf Lechtenfeld, Gabriel Picanço Costa, and Jona Röhrig. Yang–Mills solutions on Minkowski space via non-compact coset spaces. *Physics Letters B*, volume 835, page 137564, 2022. DOI:10.1016/j.physletb.2022.137564.
- 2022 **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Trajectories of charged particles in knotted electromagnetic fields. *Journal of Physics A: Mathematical and Theoretical*, volume 55, page 315401, 2022. DOI:10.1088/1751-8121/ac7c49.
- 2022 Lukas Hantzko, **Kaushlendra Kumar**, and Gabriel Picanço Costa. Conserved charges for rational electromagnetic knots. *The European Physical Journal Plus*, volume 137, page 407, 2022. DOI:10.1140/epjp/s13360-022-02563-4.
- 2021 **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Instability of cosmic Yang–Mills fields. *Nuclear Physics B*, volume 973, page 115583, 2021. DOI:10.1016/j.nuclphysb.2021.115583.
- 2020 **Kaushlendra Kumar** and Olaf Lechtenfeld. On rational electromagnetic fields. *Physics Letters A*, volume 384, page 126445, 2020. DOI:10.1016/j.physleta.2020.126445.
- 2018 **Kaushlendra Kumar** and Biswajit Chakraborty. Spectral distances on the doubled Moyal plane using Dirac eigenspinors. *Physical Review D*, volume 97, page 086019, 2018. DOI:10.1103/PhysRevD.97.086019.
- 2018 Yendrembam Chaoba Devi, **Kaushlendra Kumar**, Biswajit Chakraborty, and Frederik G. Scholtz. Revisiting Connes' finite spectral distance on noncommutative spaces: Moyal plane and fuzzy sphere. *International Journal of Geometric Methods in Modern Physics*, volume 15, page 1850204, 2018. DOI:10.1142/S0219887818502043.
- 2015 **Kaushlendra Kumar**, Shivraj Prajapat, and Biswajit Chakraborty. On the role of Schwinger's $SU(2)$ generators for simple harmonic oscillator in 2D Moyal plane. *The European Physical Journal Plus*, volume 130, page 120, 2015. DOI:10.1140/epjp/i2015-15120-7.

(b) Conference proceedings

- 2023 **Kaushlendra Kumar**. $SO(1,3)$ Yang–Mills solutions on Minkowski space via cosets. *SciPost Physics Proceedings*, volume 14, page 033, 2023. DOI:10.21468/SciPostPhysProc.14.033.
- 2023 **Kaushlendra Kumar**. On Yang–Mills fields from anti-de Sitter spaces. *Journal of Physics: Conference Series*, volume 2667, page 012018, 2023. DOI:10.1088/1742-6596/2667/1/012018.

(c) Preprints

- 2026 **Kaushlendra Kumar** and Shahn Majid. Geodesic flows on a black-hole background. *arXiv preprint*, 2026. arXiv:2603.03222 [gr-qc], preprint.

Teaching

- Spring, 2020 **Introduction to Quantum Theory**, *intensive 1-week revision course*, ITP-LUH.
- Fall, 2019 **Mathematical Methods of Physics**, *intensive 1-week revision course*, ITP-LUH.
- Spring, 2019 **Lie groups, Algebra and their Representations**, *teaching assistant*, ITP-LUH.
- Fall, 2017 **Advanced Mathematical Methods of Physics**, *teaching assistant*, IISER-K.

Spring, 2017 **Intermediate Electricity and Magnetism**, *teaching assistant*, IISER-K.

Talks

- 9 Dec, 2025 **Quantum Algebras seminar, Queen Mary University of London**, Geodesic flows on a black-hole background.
- 28 May, 2024 **Quantum Algebras seminar, Queen Mary University of London**, Octonions and the Standard Model.
- 5 Mar, 2024 **Quantum Algebras seminar, Queen Mary University of London**, Emergence of modified Einstein equation from 1-loop result in IKKT matrix model.
- 18 Aug, 2023 **Balachandran's room 316 meeting (online)**, Emergent gravity from the 1-loop effective action of the IKKT matrix model.
- 27 Jul, 2023 **QTS12 conference, Prague**, Exact gauge fields from anti-de Sitter space.
- 28 Jun, 2023 **Lechtenfeld's group seminar, Hannover**, Emergent gravity from the 1-loop effective action of the IKKT matrix model.
- 4 May, 2023 **ESI talk, Vienna**, Exact gauge fields from anti-de Sitter space.
- 15 Mar, 2023 **Solitons@Work (online)**, Exact gauge fields from anti-de Sitter space.
- 23 Aug, 2022 **String Theory Journal Club DESY, Hamburg**, Yang–Mills solutions on Minkowski space via non-compact coset spaces.
- 18 Jul, 2022 **GROUP34 conference, Strasbourg**, Yang–Mills solutions on Minkowski space via non-compact coset spaces.
- 16 Mar, 2022 **Solitons@Work (online)**, Yang–Mills solutions on Minkowski space via non-compact coset space.
- 1 Jun, 2022 **Lechtenfeld's group seminar, Hannover**, U(1) gauge theory in noncommutative geometry, pedagogical talk based on Dungen & Suijlekom, *Rev.Math.Phys* **24** (2012) 1230004.
- 20 Apr, 2022 **Lechtenfeld's group seminar, Hannover**, Spectral triple with real structure on fuzzy sphere, review talk based on Chakraborty et. al., *J.Math.Phys.* **63** (2022) 023504.
- 1 Dec, 2021 **Lechtenfeld's group seminar, Hannover**, New Yang–Mills solution on positive light-cone.
- 7 Jul, 2017 **S.N. Bose Center, Kolkata**, Spectral distances on doubled Moyal plane.
- 4 Nov, 2016 **DPS journal club IISER, Kolkata**, Angular momentum & simple harmonic oscillator in noncommutative Moyal plane.
- 11 Jul, 2016 **S.N. Bose Center, Kolkata**, Connes spectral distance on noncommutative spaces: fuzzy sphere & double Moyal plane.

Schools/Conferences

- 07–11 Jul, 2025 **A Quantum of Noncommutativity**, Workshop/conference in Ambleside, United Kingdom.
- 24–28 Jul, 2023 **QTS12**, The 12th International Symposium on Quantum Theory and Symmetries at Prague, Czech Republic.
- 06–17 Feb, 2023 **Vort2023**, Vortex Moduli workshop at ICTS, Bangalore, India.
online participation
- 29 Aug–9 Sep, 2022 **Saalburg28**, The 28th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Modern methods in quantum field theory" at Bayrischzell, Germany.
- 18–22 Jul, 2022 **GROUP34**, The 34th International Colloquium on Group Theoretical Methods in Physics at Strasbourg, France.
- 30 Aug–10 Sep, 2021 **Saalburg27**, The 27th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Gravitational waves and Black holes" at Heigenbruecken, Germany.

- 02–07 Aug, 2021 **ICMP**, International Congress on Mathematical Physics preceded by Young Researchers Symposium at Geneva, Switzerland.
- 31 Aug–11 Sep, 2020 **Saalburg26**, The 26th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "QFT, Integrable systems and String field theory" at Heigenbruecken, Germany.
- 23–27 Sep, 2019 **SHGP**, Summer School on the topic of "Special Holonomy – Geometry and Physics" at Freiburg, Germany.
- 02–13 Sep, 2019 **Saalburg25**, The 25th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Asymptotic symmetries and Hamiltonian formulation of Gravity" at Heigenbruecken, Germany.
- 15–19 Jan, 2018 **QGNCG**, Conference on "Quantum Groups & Noncommutative Geometry" at National Institute of Science Education & Research (NISER), Bhubaneswar, India.
- 06–24 Nov, 2017 **GGD2018**, Preschool + Advanced school on the topic of "Geometry, Groups and Dynamics" at International Centre for Theoretical Sciences (ICTS) Bangalore, India.

Fellowships

- 01 Mar, 2024 – 14 Nov, 2027 **Deutsche Forschungsgemeinschaft (DFG)**, Walter-Benjamin fellowship to work with Shahn Majid at Queen Mary University of London on particle physics from Quantum Riemannian Geometry.
- 01 Oct, 2018– 31 Aug, 2022 **Deutscher Akademischer Austauschdienst (DAAD)**, Scholarship to carry out (mostly doctoral) research work at ITP-LUH.
- 17 Oct, 2016– 31 May, 2018 **Council for Scientific and Industrial Research (CSIR)**, Junior Research Fellowship (JRF) to carry out research work at IISER-K.

Language skills

- Indic Maithili, Hindi and Bengali.
- English Fluent.
- German Intermediate level (B1).

Computer skills

- Languages C, HTML, CSS.
- Applications $\text{\LaTeX}$, Mathematica, GnuPlot, Amber, Matlab.
- Systems Linux, Mac, Windows.